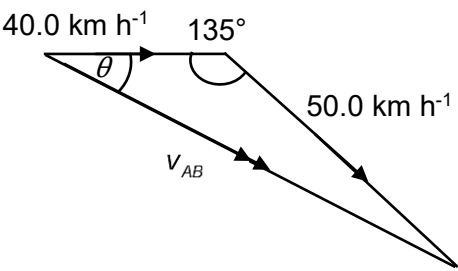


Q2		
(a)	$F_d = \frac{1}{2} \rho C_d A v^2$ $= \frac{1}{2} (1.20) (0.30) (2.50) \left(\frac{108 \times 1000}{60 \times 60} \right)^2$ $= 405 \text{ N}$ $\frac{\Delta F_d}{F_d} = \frac{\Delta \rho}{\rho} + \frac{\Delta C_d}{C_d} + \frac{\Delta A}{A} + \frac{2 \Delta v}{v}$ $\frac{\Delta F_d}{405} = \frac{0.05}{1.20} + \frac{0.02}{0.30} + \frac{0.05}{2.50} + \frac{2(2)}{108}$ $\Delta F_d = 70 \text{ N (1s.f.)}$ $F_d \pm \Delta F_d = 410 \pm 70 \text{ N}$	
(b)	velocity of car A relative to car B, $v_{AB} = v_A - v_B$  By Cosine Rule, $v_{AB} = \sqrt{40.0^2 + 50.0^2 - 2(40.0)(50.0)\cos 135^\circ}$ $= 83.2 \text{ km h}^{-1}$ By Sine Rule, $\frac{\sin \theta}{50.0} = \frac{\sin 135^\circ}{83.2}$, hence $\theta = 25.1^\circ$ direction: bearing = $90.0^\circ + 25.1^\circ = 115.1^\circ$	

